MATH 2241 INTRODUCTION TO MATHEMATICAL ANALYSIS
FALL SEMESTER 2026
KEY POINTS FOR CHAPTERS 1 AND 2

Chapter 1: Real Numbers

(1) Examples and non-examples of ordered fields.
(2) How to deduce some properties of fields, or ordered fields, from given axioms. You do not have to memorize the axioms; they will be given if relevant questions are asked.
(3) Definitions of supremum and infimum. The relationship between supremum and infimum.
(4) How to find the supremum and infimum of some sets. How to find the supremum and infimum, if possible, under some operations, e.g., addition and multiplication.
(5) The completeness property and how to use this axiom to show the existence of a solution to an equation.
(6) The definition of the real numbers.
(7) Explain why $\mathbb{Q}$ fails to satisfy the completeness property.
(8) The well-ordering property of $\mathbb{N}$.
(9) The density property of the set of rational numbers.
(10) State and prove the Nested Interval Property. Examples of nested intervals: the empty set, one element, more than one element, etc.
(11) How to show that $\mathbb{Q}$ is countable.
(12) How to show that $\mathbb{R}$, or the set of irrational numbers, is uncountable.
(13) How to show that a number, e.g., $\sqrt{2}$, is not rational.

Chapter 2: Sequences

(1) Definition of limits, i.e., the ϵ-N definition. Use the ϵ-N definition to find some limits.
(2) Examples of convergent and divergent sequences. Operations on convergent and divergent sequences.
(3) Methods for showing that a sequence is convergent: boundedness and monotonicity, the Monotone Convergence Theorem, Cauchy sequences, the squeeze theorem, and the use of known limits, e.g.,

$$\lim_{n\to\infty} \sqrt[n]{n}, \quad \lim_{n\to\infty} \left(1+\frac{1}{n}\right)^n, \quad \lim_{n\to\infty} r^n, \quad \lim_{n\to\infty} \frac{\sin(n)}{n}.$$

(4) How to prove the Monotone Convergence Theorem from the definitions.
(5) How to use the Monotone Convergence Theorem to find limits of some recursive relations.

(6) Definition of Cauchy sequences. Their relationship to convergent sequences. How to show that a convergent sequence is Cauchy, and how to prove the converse.
(7) Contractive sequences are Cauchy, and examples.
(8) The subsequence criterion for convergence of a sequence.
(9) State the Monotone Subsequence Theorem and the Bolzano-Weierstrass Theorem. How to use the Monotone Subsequence Theorem to prove the Bolzano-Weierstrass Theorem. Find monotone subsequences of given sequences.
(10) Definition of a series. Examples of convergent series.
(11) Absolute and conditional convergence, and examples. Relationships among absolute convergence, conditional convergence, and convergence, and how to prove these relationships.
(12) Rearrangements of series, e.g., for absolutely convergent series and conditionally convergent series. Examples of rearrangements of series. The Rearrangement Theorem.
(13) Tests for convergence: the nth-term test, comparison tests, and limit comparison tests.